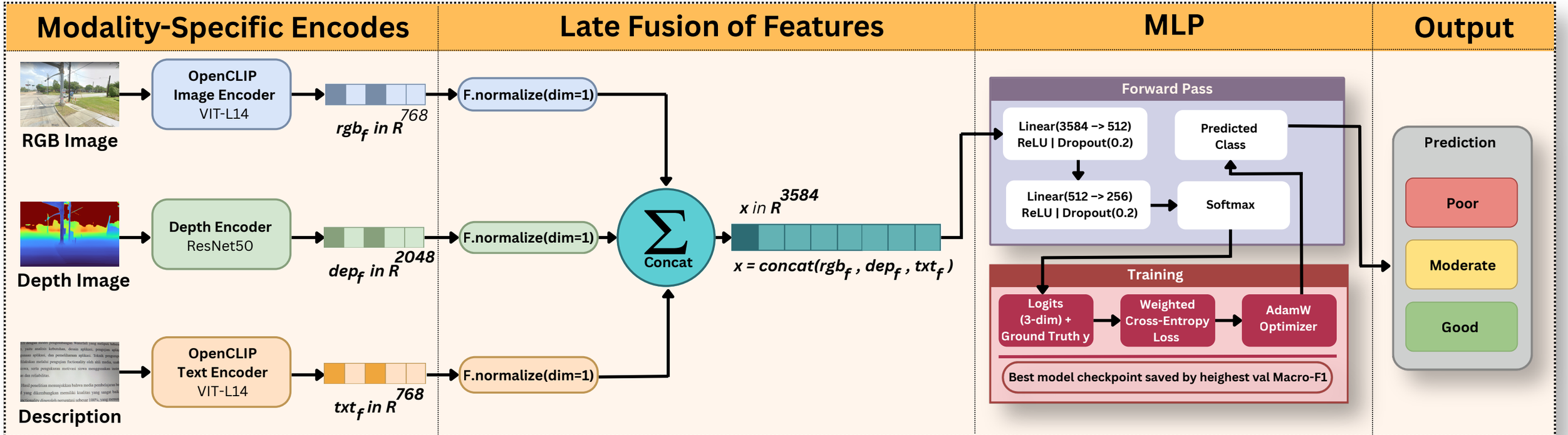
Modality-Specific Encodes
Late Fusion of Features
MLP
Output
RGB Image
OpenCLIP Image Encoder VIT-L14
rgb_f in R^{768}
F.normalize(dim=1)
Depth Image
Depth Encoder ResNet50
dep_f in R^{2048}
F.normalize(dim=1)
Description
OpenCLIP Text Encoder VIT-L14
txt_f in R^{768}
F.normalize(dim=1)
Concat
x in R^{3584}
$x = concat(rgb_f, dep_f, txt_f)$
Forward Pass
Linear(3584 –> 512) ReLU | Dropout(0.2)
Linear(512 –> 256) ReLU | Dropout(0.2)
Softmax
Predicted Class
Training
Logits (3-dim) + Ground Truth y
Weighted Cross-Entropy Loss
AdamW Optimizer
Best model checkpoint saved by heighest val Macro-F1
Prediction
Poor
Moderate
Good